ESIP SUMMER MEETING 2026 • AUSTIN, TEXAS

# CODED - Cloud-Optimized Distributed Earth Data

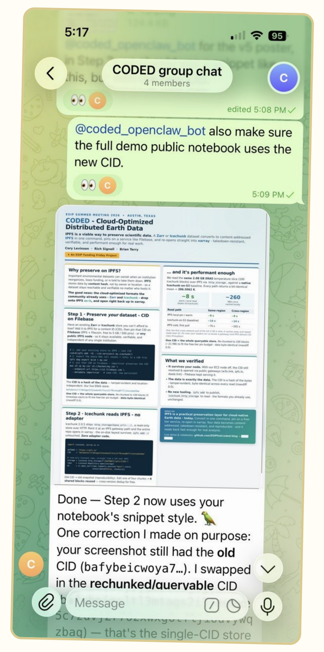
**IPFS (InterPlanetary File System) is a viable way to preserve scientific data.** A **Zarr** or **Icechunk** dataset converts to content-addressed **IPFS** in one command, pins on a service like Filebase, and re-opens straight into **xarray** - takedown-resistant, verifiable, and performant enough for real work.

**Cory Levinson · Rich Signell · Brian Terry**

**★ An ESIP Funding Friday Project**

**How we built this →**

**We chatted with an OpenClaw AI agent over Telegram - it ran the benchmarks, spun nodes up and down, and drafted these figures.**

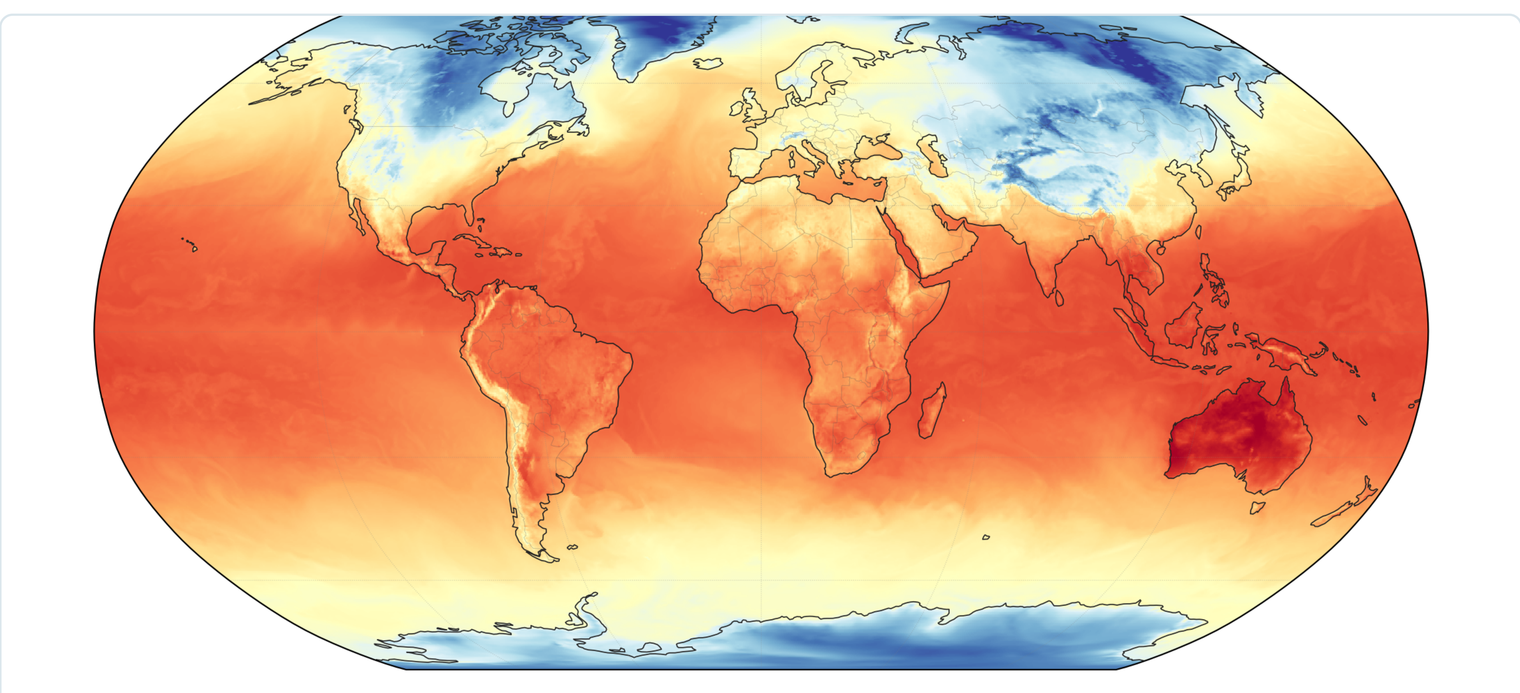


## Why preserve on IPFS?

Important environmental datasets can vanish when an institution reorganizes, loses funding, or is told to take them down. **IPFS** stores data by **content hash**, not by owner or location - so a dataset stays reachable and verifiable no matter who hosts it.

**The good news: the cloud-optimized formats the community already uses - Zarr and Icechunk - drop onto IPFS as-is, and open right back up in xarray.**

## Live proof · ERA5 straight from IPFS



**Live proof:** this global ERA5 2 m-temperature field was opened **straight from IPFS** - `icechunk.http_storage("ipfs.io/ipfs/<CID>")` → `xr.open_zarr` - in **<1 s**. Content-addressed data, real analysis, no download step.

## Step 1 · Preserve your dataset - CID on Filebase

Have an existing **Zarr** or **Icechunk** store you can't afford to lose? Add it to IPFS for a content identifier (CID), then pin that CID on **Filebase** (an IPFS + Filecoin pinning service, free to 5 GB / 500 pins) - or **any public IPFS node** - so it stays available, verifiable, and independent of any single institution.

```
# add your existing store to IPFS → one root CID
cid=$(ipfs add -rQ --cid-version=1 my.icechunk/)
# pin that CID on Filebase (IPFS + Filecoin) — stays alive
# independent of your own node. Same CID, now persistent.
```

The **CID is a hash of the data** — tamper-evident and location-independent. Our live ERA5 store:

`bafybeiecltl3mtags2i3jeumxe5c7zuvj2r76zxwxg6tfljiouvywqzbaq`

**One CID = the whole queryable store.** Re-chunked to 100 blocks (5 timesteps each) to fit the free-tier pin budget - **data byte-identical** (maxdiff 0.0).

## Step 2 · Icechunk reads IPFS - no adapter

Icechunk 2.0.5 ships `http_storage(base_url=...)`, a read-only store over HTTP. Point it at an IPFS gateway path and the whole repo opens in xarray - the on-disk layout survives `ipfs add -r` untouched. **Zero adapter code.**

```python
import icechunk, xarray as xr

GATEWAY = "https://ipfs.io"
CID     = "bafybeiecltl3mtags2i3jeumxe5c7zuvj2r76zxwxg6tfljiouvywqzbaq"

# read-only Icechunk repo, straight from a CID over IPFS
storage = icechunk.http_storage(f"{GATEWAY}/ipfs/{CID}")
repo    = icechunk.Repository.open(storage)
ds      = xr.open_zarr(repo.readonly_session("main").store,
                       consolidated=False, chunks={})
```

Old CID → old snapshot (reproducibility). Dedup across versions comes free - edits reuse unchanged blocks.

## … and it's performant enough

We read the **same 2.08 GB ERA5** temperature slice (100 Icechunk blocks) over IPFS via `http_storage`, against a **native Icechunk-on-S3** baseline. Every path returns a bit-identical mean = **286.5062 K**.

**~8 s**
warm / local read
(beats S3-Icechunk)

**~260**
MB/s once
cached locally

| Read path | Same-region | Cross-region |
|---|---|---|
| IPFS local pin / warm | **~8 s** | **~8 s** |
| Icechunk-on-S3 (baseline) | **~14 s** | **~14 s** |
| IPFS cold, first pull | ~75 s | ~101 s |

Only the first cross-network pull of the full 2 GB is slow; it caches away and repeat reads beat S3. Pin near your readers (or co-locate a gateway) and IPFS delivers S3-class throughput.

## What we verified

- **It survives your node.** With our EC2 node off, the CID still resolved & opened via public gateways (Filebase, ipfs.io, dweb.link) - Filebase kept serving it.
- **The data is exactly the data.** The CID is a hash of the bytes - tamper-evident, byte-identical across every read (maxdiff 0.0).
- **No new tooling.** `ipfs add` to publish, `icechunk.http_storage` to read - the formats you already use, unchanged.

## Bottom Line

**IPFS is a practical preservation layer for cloud-native Earth data - today.** Convert in one command, pin it on a cooperating organization's IPFS node or a commercial pinning service, re-open in xarray. Your data becomes content-addressed, takedown-resistant, and reproducible - and it reads back fast enough for real analysis.

Live demo & notebooks: **github.com/ESIPFed/coded-blog** (see **ipfs-agent/**)

**Scan the codes below to run it yourself →**

**Acknowledgements:** This work was carried out by **chatting with an OpenClaw AI agent over Telegram** (AWS VM, Claude via Amazon Bedrock) - the team steered; the agent ran the benchmarks, provisioned & tore down nodes, and drafted the figures. Thanks to AWS for the ESIP Credits!